# How Einstein Induced the Photoelectric Effect: Machine Learning

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# The Central Claim

## Thesis

When Einstein proposed the light-quantum hypothesis in 1905 to explain the photoelectric effect, he was—in the language we now use—doing **machine learning** and **statistical inference**.

**Three acts:**

1. **Theory** (1905): Einstein proposes the model that defines the hypothesis class
2. **Data quality** (1887–1916): experimenters spend 29 years collecting clean training data
3. **Estimation** (1916): Millikan runs the regression and recovers Planck's constant to 0.5%

The episode illustrates *both* the power of theory-guided learning *and* the limits of purely data-driven AI for scientific discovery.

# Background: The Photoelectric Effect

**The phenomenon (Hertz, 1887; Hallwachs, 1888):**

- Shine light on a metal surface
- Electrons are ejected
- Apply a *stopping potential* $V_0$ to just halt the fastest electrons

**The puzzle (Lenard, 1902):**

- $V_0$ rises with light *frequency* $\nu$
- $V_0$ is **independent** of light *intensity* $I$
- Classical wave theory predicted the opposite

## Classical Prediction (Wrong)

Wave amplitude $A \propto \sqrt{I}$ delivers energy

$\Rightarrow V_0$ should rise with $I$

$\Rightarrow V_0$ should be independent of $\nu$

## What Lenard Found

Exactly the opposite!

Intensity $I$: **irrelevant**

Frequency $\nu$: **everything**

# Lenard's Finding as Feature Engineering

**A modern ML framing of Lenard (1902):**

Lenard ran an ablation experiment. He measured $V_0$ as a function of *both* $\nu$ and $I$.

| Feature | Effect on $V_0$ | Importance |
|---|---|---|
| Frequency $\nu$ | Strong $\uparrow$ | **High** |
| Intensity $I$ | None | **Zero** |
| Wavelength $\lambda$ | (same as $\nu$) | High |

**Conclusion:** drop $I$ from the feature set; keep $\nu$.

## ML Translation

**Input features:** $\nu$, $I$, $\lambda$, ...

**Label:** stopping potential $V_0$

**Lenard's result:** feature-importance analysis eliminates $I$; selects $\nu$ as the *sole relevant input*

"Lenard ... established **two qualitative facts**: $V_0$ is independent of intensity; $V_0$ rises with frequency."

— Millikan (1916)

# Einstein's 1905 Insight: Model Architecture

**The quantum hypothesis:**

Each light quantum carries energy $E = h\nu$.

A single quantum is absorbed by a single electron.

If $h\nu > W$ (work function), the electron escapes.

**Energy conservation** gives:

$$\underbrace{eV_0}_{\text{max KE}} = \underbrace{h\nu}_{\text{photon energy}} - \underbrace{W}_{\text{work function}}$$

## The Prediction (1905)

$$\boxed{V_0 = \frac{h}{e}\nu - \frac{W}{e}}$$

Linear in frequency.

Slope $= h/e$ is **universal** across all metals.

Intercept $= -W/e$ is metal-specific.

**Written before a single reliable $V_0$-vs-$\nu$ measurement existed!**

## The Inductive Bias

The quantum hypothesis restricts the hypothesis class from *all functions* of $(\nu, I, \lambda, \ldots)$ to a **single affine function in $\nu$ alone**, with a specific universal slope.

# The Regression Equation

**Adding measurement noise and a nuisance parameter:**

$$V_{0i} = \underbrace{\alpha}_{\text{intercept}} + \underbrace{\beta}_{h/e}\,\nu_i + \underbrace{\varepsilon_i}_{\text{noise}}, \qquad \alpha = -\frac{W}{e} + \underbrace{v_c}_{\text{contact EMF}}$$

## Parameters of Interest

| | |
|---|---|
| $\beta = h/e$ | slope $\Rightarrow$ Planck's $h$ |
| $\alpha$ | intercept (metal-specific) |

## Nuisance Parameter

$v_c$: EMF (electromotive force) between emitter and collector. Can be as large as 3 V—comparable to $V_0$ itself!

Shifts intercept; **does not bias slope** if constant within a run.

## The Supervised Learning Problem

**Training data:** $\{(\nu_i, V_{0i})\}_{i=1}^{n}$

**Task:** estimate $\beta = h/e$

**Loss:** squared error

**Hypothesis class:** affine functions of $\nu$

**Inductive bias:** $\beta > 0$; universal across metals

# 29 Years of Collecting Training Data (1887–1916)

| Experimenter | Year | Data quality | Problem |
|---|---|---|---|
| Hertz / Hallwachs | 1887–88 | Qualitative only | No labeled $(\nu, V_0)$ pairs |
| J. J. Thomson | 1899 | Identifies electrons | No $V_0$-vs-$\nu$ measurement |
| Lenard | 1902 | Qualitative labels | No reliable numerical $V_0$ |
| Ladenburg | 1907 | 60 nm range | Linear $\approx$ square-root |
| Hughes | 1912 | 3 UV lines | Contact EMF uncontrolled; 65% scatter in $h$ |
| Richardson & Compton | 1912 | 8 metals | Same problems; 60% scatter |
| **Millikan** | **1916** | **Clean surfaces, wide $\Delta\nu$, controlled $v_c$** | **$h$ to 0.5%** |

## Data Quality is the Binding Constraint

The regression is trivial once you have clean data. The 29-year delay reflects *data engineering*, not statistical sophistication.

# Millikan's Experimental Design (1916)

**Key data-quality innovations ("a machine shop in vacuo"):**

- ✓ **Fresh surfaces:** electromagnetic knife shaved the alkali-metal cylinder in vacuum—no oxidation, no work-function contamination
- ✓ **Spectrally pure lines:** mercury arc with filters to exclude UV companions—eliminates stray-light bias on $V_0$
- ✓ **Wide frequency range:** 5461 Å to 2535 Å ($\Delta\nu = 6.3 \times 10^{14}$ Hz, nearly a factor of two)
- ✓ **Controlled nuisance:** Kelvin double-balance measured $v_c$ continuously; verified stable to $\pm 0.01$ V over 9 hours

## Why the Wide Frequency Range Matters (Cramér-Rao)

$$\mathrm{SE}(\hat{\beta}) = \frac{\sigma}{\sqrt{S_{\nu\nu}}}, \quad S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$$

Wider $\Delta\nu \Rightarrow$ larger $S_{\nu\nu} \Rightarrow$ smaller SE. Hughes's 60 nm range gives a standard error **2.2× larger** than Millikan's full range.

# Millikan's Training Data

| $\lambda$ (Å) | $\nu$ ($10^{14}$ Hz) | $V_0^{\mathrm{Na}}$ (V) | $V_0^{\mathrm{Li}}$ (V) |
|---|---|---|---|
| 5461.0 | 5.490 | 0.454 | — |
| 4339.4 | 6.908 | 1.039 | 0.499 |
| 4046.8 | 7.409 | 1.246 | 0.706 |
| 3650.2 | 8.213 | 1.578 | 1.037 |
| 3125.5 | 9.591 | 2.147 | 1.606 |
| 2534.7 | 11.827 | 3.030 | 2.529 |

True stopping potential $= V_{0,\mathrm{obs}} + v_c$ (Na: $v_c = 2.51\,\mathrm{V}$; Li: $v_c = 1.52\,\mathrm{V}$)

- 6 labeled training examples per metal
- Two metals (sodium and lithium) available
- Points span nearly a **factor of two** in frequency

# OLS Regression: Estimating Planck's Constant

**Primary regression:** sodium, five points (omitting 2535 Å scattered-light correction)

**Standard OLS formula:**

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{3.830}{9.277} = \mathbf{4.129 \times 10^{-15}}\ \mathrm{V{\cdot}s}$$

$$\hat{\alpha} = \bar{V}_0 - \hat{\beta}\,\bar{\nu} = -1.813\,\mathrm{V}$$

| Statistic | Value |
|---|---|
| $R^2$ | $\approx 0.99999$ |
| Residuals | $\leq 2\,\mathrm{mV}$ |
| SE($\hat{\beta}$) | $4.6 \times 10^{-18}$ V·s |
| Rel. precision | **0.11%** |

**Implied Planck's constant:**

| *e* used | Implied *h* |
|---|---|
| Millikan's *e* (1916) | $6.573 \times 10^{-34}$ |
| Modern NIST *e* | $6.614 \times 10^{-34}$ |
| **Modern exact** | $\mathbf{6.626 \times 10^{-34}}$ |

## Result

*h* recovered to **0.5%** of its modern value from **5 data points**.

# MLE = OLS Under Gaussian Errors

**Parametric model:**

$$V_{0i} = \alpha + \beta\,\nu_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$$

**Log-likelihood:**

$$\ell(\alpha, \beta, \sigma) = -\frac{n}{2}\ln(2\pi) - n\ln\sigma - \frac{1}{2\sigma^2}\underbrace{\sum_i (V_{0i} - \alpha - \beta\nu_i)^2}_{\mathrm{RSS}(\alpha,\beta)}$$

Maximising $\ell$ over $(\alpha, \beta)$ minimises RSS—exactly OLS.

$$\boxed{(\hat{\alpha}_{\mathrm{MLE}}, \hat{\beta}_{\mathrm{MLE}}) = (\hat{\alpha}_{\mathrm{OLS}}, \hat{\beta}_{\mathrm{OLS}})}$$

## Implication

Every OLS result carries over to MLE. The MSE training loss used in neural-network regression *is* the Gaussian log-likelihood—the physics and the ML objective coincide.

# The Cramér-Rao Bound and Experimental Design

**Fisher information and Cramér-Rao bound for $\beta = h/e$:**

$$\mathcal{I}_{\beta\beta} = \frac{S_{\nu\nu}}{\sigma^2}, \qquad \mathrm{Var}(\tilde{\beta}) \geq \frac{\sigma^2}{S_{\nu\nu}}, \quad S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$$

OLS achieves this bound (Gauss-Markov + Cramér-Rao).

| Dataset | $n$ | $S_{\nu\nu}$ ($10^{28}$ Hz$^2$) | SE($\hat{\beta}$) ($10^{-18}$ V·s) |
|---|---|---|---|
| Restricted (4047–2535 Å, like Hughes) | 3 | 1.96 | 10.0 |
| Millikan 5-pt (5461–3126 Å) | 5 | 9.28 | 4.6 |
| Millikan full (5461–2535 Å) | 6 | 28.14 | 2.6 |

## Design lesson

Span frequencies widely. The CR bound justifies Millikan's choice of a nearly 2× range in $\nu$ over the narrow 60 nm UV band used by predecessors.

# Universality: Zero-Shot Transfer Across Metals

**The quantum hypothesis predicts:** $\beta = h/e$ is the same for *all* metals. This is a strong, falsifiable structural restriction.

**Separate regressions:**

$$\hat{\beta}_{\mathrm{Na}} = 4.129 \times 10^{-15} \text{ V·s}$$

$$\hat{\beta}_{\mathrm{Li}} = 4.137 \times 10^{-15} \text{ V·s}$$

$$\text{difference} = 0.19\%$$

Wald test of $H_0 : \beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$:

$$z \approx 0.87 \quad (\ll 1.96)$$

Cannot reject universality.

## ML Language

**Pre-training:** fit slope on sodium

**Zero-shot transfer:** apply slope to lithium (different intercept only)

**Fine-tuning needed:** only intercept $\alpha_{\mathrm{Li}}$ (work function shifts)

**Transfer error:** 0.2%

The pooled estimate $\hat{\beta}_{\mathrm{pool}} \approx 4.133 \times 10^{-15}$ V·s has SE reduced by $\approx 1/\sqrt{2}$ relative to either single-metal estimate.

# Historical Estimates of Planck's Constant

| Authors | Year | $h$ ($10^{-27}$ erg·s) | Error |
|---|---|---|---|
| Hughes | 1912 | 3.55–5.85 | −12 to −46% |
| Richardson & Compton | 1912 | 3.5–5.8 | −12 to −47% |
| Planck (black body) | 1913 | 6.415 | −3.2% |
| Millikan (Na, 5-pt) | 1916 | 6.573 | −0.80% |
| Millikan (Na, 9-pt) | 1916 | 6.578 | −0.73% |
| Millikan (Li) | 1916 | 6.584 | −0.64% |
| **Millikan mean** | **1916** | **6.579** | **−0.72%** |
| Modern NIST 2022 | 2022 | 6.6261 | 0 |

**Pattern:** pre-Millikan estimates biased downward 10–47%.

Three causes: stray UV (inflates $V_0$ at long wavelengths $\Rightarrow$ flattens slope), narrow $\Delta\nu$ (large SE), uncontrolled $v_c$ (slope bias).

Millikan solved all three.

# Millikan's Epistemic Reluctance

Millikan confirmed Einstein's equation to 0.5%. **Then wrote:**

> *"Despite then the apparently complete success of the Einstein equation, the physical theory of which it was designed to be the symbolic expression is found* **so untenable** *that Einstein himself, I believe, no longer holds to it."*
>
> — Millikan (1916), p. 388

He had hoped the experiment would **refute** the equation!

## Lesson from Kepler and Newton

Einstein recovered a first-class "Kepler stage" model in sense of Koopmans (1947)

Millikan confirmed the slope $h/e$ to 0.5% without accepting that light consists of quanta.

Compton scattering (1923) finally convinced him.

Nobel acceptance (1923):
"...contrary to my own expectation, **first direct experimental proof** ...of the complete validity of the Einstein equation."

# The ML-to-Physics Dictionary

| Scientific step | ML name | Who |
|---|---|---|
| Propose $E = h\nu$; derive $V_0 = (h/e)\nu - W/e$ | Inductive bias + Hypothesis class | Einstein (1905) |
| $V_0$ depends on $\nu$ not $I$ | Feature engineering | Lenard (1902) |
| Measure $(\nu_i, V_{0i})$ pairs | Labeled training data | Millikan (1916) |
| Control contact EMF $v_c$ | Nuisance-variable identification | Millikan (1916) |
| Min. $\sum(V_{0i} - \hat{V}_{0i})^2$ | MSE training (OLS/MLE) | Millikan; this paper |
| $\lvert\hat{\beta} - h/e\rvert < 1\%$ | Test accuracy / Confirmation | Millikan (1916) |
| Na and Li give same $\hat{\beta}$ | Zero-shot transfer learning | Millikan (1916) |

# The Hypothesis Class: Where Einstein Differs from atheoretical ML

**Einstein's hypothesis class:**

$$\mathcal{H} = \left\{ f_{\alpha,\beta} : \nu \mapsto \alpha + \beta\nu \;\middle|\; \alpha \in \mathbb{R},\ \beta > 0 \right\}$$

with the *cross-metal restriction* that $\beta$ is universal.

- VC dimension $= 2$
- One-parameter family of parallel lines
- Derived *deductively* from $E = h\nu$
- Five training points are *more than sufficient*

## What atheoretical ML Misses

A linear regression package gives the right numerical answer—but:

1. Cannot identify $\hat{\beta} = 4.129 \times 10^{-15}$ as $h/e$
2. Cannot impose $\beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$ *from first principles*
3. Cannot decompose intercept into $W/e$ vs. $v_c$
4. Cannot reduce data requirement from $O(10^3)$ to 5

## Parameter Space Reduction Einstein Achieved

$E = h\nu$ reduced the hypothesis space from the infinite-dimensional function class to a **two-dimensional manifold of parallel lines**— compressing the data requirement by three orders of magnitude.

# Einstein vs. a Deep Neural Network: Three Epochs

| Epoch | $n$ avail. | Einstein model | DNN (VC $\approx$ 5700) | DNN outcome |
|---|---|---|---|---|
| **Lenard 1902** | $\approx 0$ | Sufficient$^\dagger$ | Insufficient | Cannot train (no labels) |
| **Hughes 1912** | $\sim 30$ | Sufficient | Insufficient ($\times 190$) | Memorises; no generalisation |
| **Millikan 1916** | 5 | Sufficient | Very insufficient | Interpolates; no inference |
| **Modern era** | $\sim 5000$ | Sufficient | Sufficient | Recovers line; cannot identify $h/e$ |

$^\dagger$Functional form derived from theory; only two free parameters remain.

A three-layer DNN (16 neurons/layer) has $p \approx 593$ parameters and VC dim $\approx 5700$. Reliable 1% generalisation requires $n \gtrsim 5700$—short by three orders of magnitude at Millikan's epoch.

# The DNN on Millikan's Five Points

**What happens if you train any DNN on $n = 5$ data points?**

**Interpolation regime** (any DNN with $p > 5$):

- Training loss $= 0$ (memorises all points)
- Function is unconstrained *between* points
- Cannot estimate slope uncertainty: zero residual df

**OLS with $n = 5$, $p = 2$:**

- 3 residual degrees of freedom
- $\mathrm{SE}(\hat{\beta}) = 4.6 \times 10^{-18}$ V·s
- 0.11% relative precision, at the Cramér-Rao bound

## The Core Lesson

**Einstein's theory** (not more data, not a better algorithm) is what made five training points sufficient.

Specifying the hypothesis class was a harder intellectual act than fitting the line.

## Where DNNs Prevail

High-dimensional problems where no parsimonious parametric family is known (images, proteins, language).

The photoelectric problem sits at the *opposite* extreme.

# A Broader Lesson: The Hypothetis-Deduction Loop

| Step | Science | ML |
|---|---|---|
| Propose $E = h\nu$ | Theoretical hypothesis | Model architecture |
| Derive $V_0 = (h/e)\nu - W/e$ | Deduction | Hypothesis class |
| Choose $\nu$, not $I$ | Experiment design | Feature engineering |
| Measure $(\nu_i, V_{0i})$ | Observation | Training data |
| Minimise RSS | Least squares | MSE training |
| $\lvert\hat{\beta} - h/e\rvert < 1\%$ | Confirmation | Test accuracy |
| Na and Li: same $\hat{\beta}$ | Universality test | Zero-shot transfer |

## Modern AI for Science — the Missing Piece

Modern systems (AI Feynman, symbolic regression) can *induce* the linear law from data by exhaustive search, but they do so over a combinatorially large function space. Einstein's deductive step encoded the result of that search in one paragraph.

**The human contribution—theory construction—remains indispensable.**

# Bonus: Born's Rule and an Unintended Consequence

Einstein's quantum hypothesis required:

$$|E|^2 \propto \text{photon } \textit{density}$$

The classical wave intensity $|E|^2$ becomes a **probability density**, not just an energy density.

Einstein called this the *Gespensterfeld* (ghost field)—a wave that guides probabilities.

**Max Born (1926)** applied Einstein's insight to interpret Schrödinger's wave function:

$$\boxed{|\psi|^2 = \text{probability density}}$$

**Born's rule** — quantum mechanics.

## Two Nobel Prizes

**1921:** Einstein — for the equation that made the measurement possible

**1954:** Born — for the probabilistic interpretation the equation made *inevitable*

Einstein spent the rest of his life rejecting Born's rule as a "complete description of reality."

He won the Nobel for a paper whose logic seeded the interpretation he most opposed.

# Summary and Contributions

## The Thesis, Revisited

Einstein was doing statistics with a tightly parameterized model in 1905. ML systems are yet to automate a key step: **specifying the hypothesis class**.

**Three principles illustrated:**

1. **Theoretical priors guide design.** Once $E = h\nu$ fixes the functional form, only 5 training points are needed for 0.1% precision.
2. **Data quality is the binding constraint.** The regression is trivial; cleaning the data took 29 years.
3. **Confirming an equation $\neq$ endorsing the model.** Millikan confirmed the slope to 0.5% while publicly rejecting the light-quantum interpretation.

## The Limit of Pure AI

A data-driven system given only $(\nu_i, V_{0i})$ pairs can recover $\hat{\beta} = 4.129 \times 10^{-15}$ V·s but cannot know it is $h/e$. **Physical meaning requires physical theory.**

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